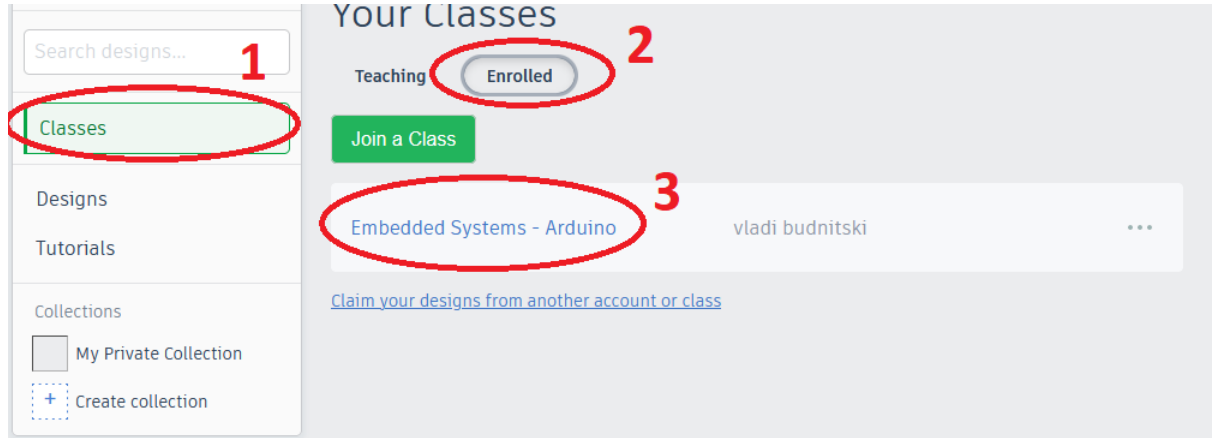


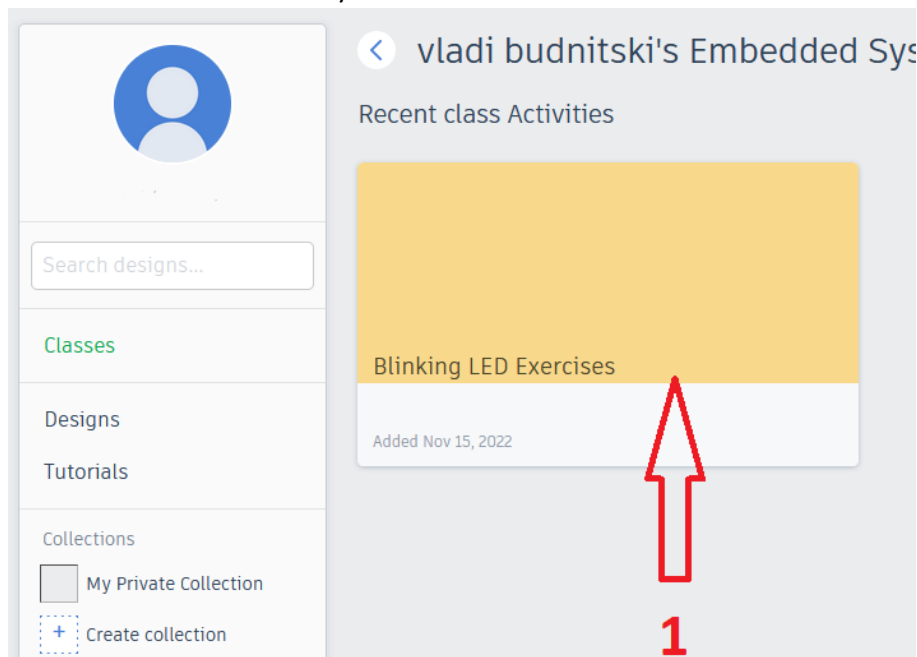
System Configuration and Exercises Submissions Understanding

After you've joined the Class:

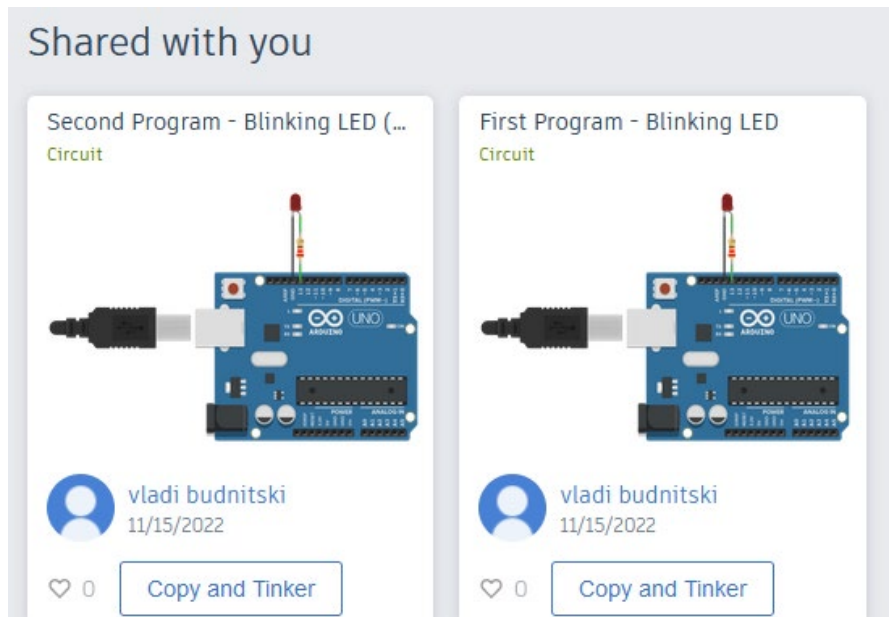
a. Navigate to:



b. Get into the relevant activity:



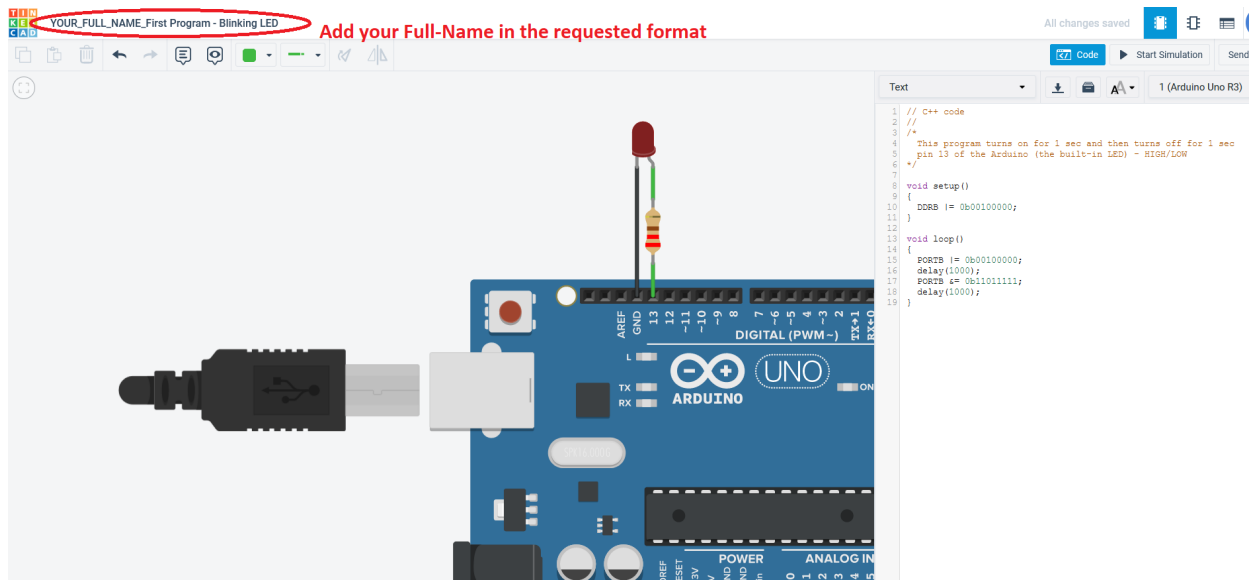
c. You can observe various designs shared with you.
For example:



- d. Every task that you will need to work on, press on “Copy and Tinker” button.

This will create a “copy” of the “pre-made” design that you can work on, edit, edit/complete the code, etc.

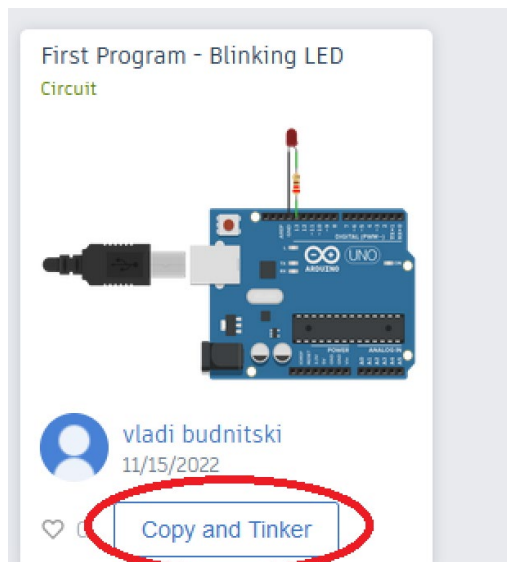
- e. Note to modify the Tinkered Design file name (so that it will be “properly submitted” and graded). For example:



Exercises

1. “First Program – Blinking LED”

- a. Go to “Activities” and “Copy and Tinker” the associated task:



- b. Update your Tinkered design’s file name (e.g., “Vladi_First Program – Blinking LED”).
- c. Observe the design, look at the code, make sure everything is clear.
- d. Run a simulation (**initial code is already there!**).
- e. **Once you make sure everything works – YOUR TASK is to make the following updates:**
- i. Set the 9 pin as OUTPUT. Make sure to make also the Hardware Connection (connecting the LED to 9).
 - ii. Blink the LED ON for 1 second(s), then OFF for 3 second(s).

2. “Second Program – Blinking LED (without "delay")”

- a. Go to “Activities” and “Copy and Tinker” the associated task:
“Second Program – Blinking LED (without "delay")”
- b. Update your “submission file name” as mentioned in the previous page (e.g., “Vladi_Second Program – Blinking LED (without "delay")”).

c. Your task:

- i. Configure pin 7 as OUTPUT.
- ii. Turn the LED ON for 1 second(s), then OFF for 1 second(s), without using the built-in “delay()” function.

Instead, use the following code snippet:

```
// fill in a guess for your CPU speed to get ~1s  
long i = ?????;  
do { i--; } while (i != 0);
```

NOTES:

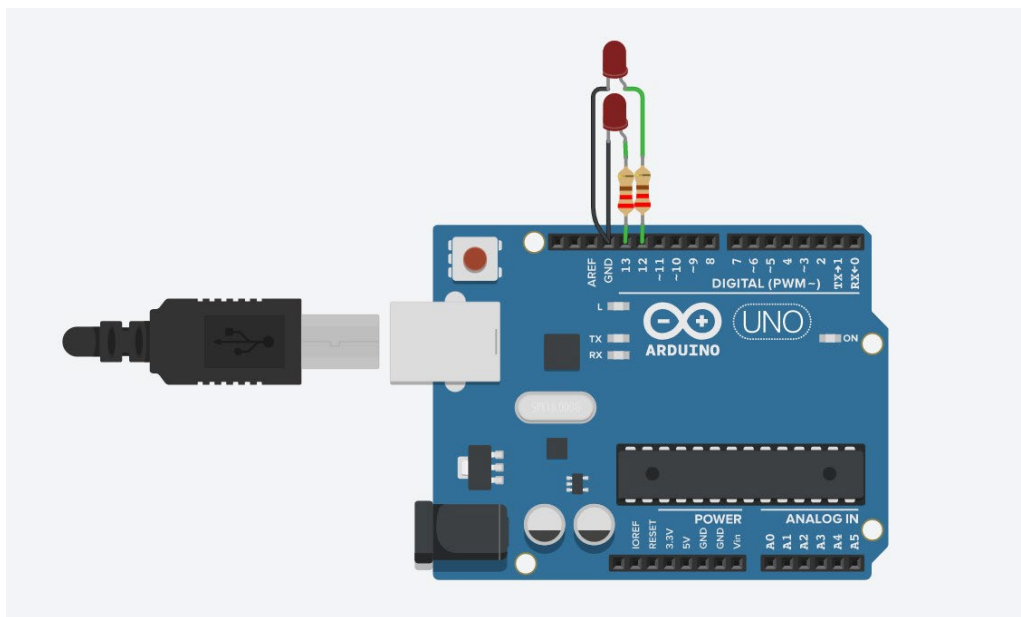
1. The times are NOT EXPECTED to be EXACT! Don’t aim for that...
2. This code is just a simple example of “using the CPU” without the built-in “delay” function.
 - a. It might give different results (and inaccurate) results during different executions – it’s OK!
 - b. Later, during our semester, we’ll learn about Timers (where we can configure EXACTLY what we need, precise timing, no CPU overload, etc) 😊

3. “Third Program - 2 Blinking LEDs – one after the other”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:
 - I. Arduino Uno r3.
 - II. 2 LEDs (any color) + 2 x 220Ω resistors
 - III. Pin connections: LED1 on pin 4, LED2 on pin 3.
- c. Your task:
 - i. Set both pins as OUTPUT.
 - ii. Blink LED1 ON for 1 second(s) while LED2 is OFF.
 - iii. Then Blink LED2 ON for 1 second(s) while LED1 is OFF.

(Make sure the pins are connected exactly as required)

(For example, in the following design the LEDs are connected to D12, D13)



NOTES:

You can use the “delay” function for this and the following exercises in this document.

4. “Fourth Program 2 Blinking LEDs – Simultaneously”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 2 LEDs (any color) + 2 x 220Ω resistors
 - iii. Pin connections: LED1 on pin 4, LED2 on pin 8.
- c. Your task:
 - i. Set both pins as OUTPUT.
 - ii. Turn both LEDs ON for 5 second(s).
 - iii. Turn both LEDs OFF for 1 second(s).

5. "Fifth Program - Traffic_Light_Simulator"

- a. Go to "Activities", "Copy and Tinker" the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 3 LEDs (Red, Yellow, Green) + 3 x 220Ω resistors.
 - iii. Pin connections: Red on 6, Yellow on 9, Green on 4.
- c. Your task:
 - i. Turn Red ON for 7 second(s).
 - ii. Then also Yellow ON for 4 second(s).
 - iii. Turn both Red and Yellow LEDs OFF and turn Green ON for 2 second(s).
 - iv. Turn Green OFF and Yellow ON for 4 second(s).
 - v. Turn Yellow OFF and Red ON for 7 second(s).
 - vi. Repeat.

(This is just an EXAMPLE. Make sure the pins are connected exactly as required!)

